

Area Between Curves

ANSWER KEY



Area between a curve and the x-axis

- 1 Find the area of the region bounded by $y = 4 - x^2$ and the x-axis.

Roots at $x = \pm 2$. Area $= \int_{-2}^2 (4 - x^2) dx = \left[4x - \frac{x^3}{3} \right]_{-2}^2 = \left(8 - \frac{8}{3} \right) - \left(-8 + \frac{8}{3} \right) = \frac{16}{3} + \frac{16}{3} = \frac{32}{3}$.

Area between two curves (vertical strips)

- 2 Find the area between $y = x^2$ and $y = x + 2$.

Intersect: $x^2 = x + 2 \Rightarrow x^2 - x - 2 = 0 \Rightarrow (x - 2)(x + 1) = 0$; $x = -1, 2$. On this interval $x + 2 \geq x^2$. Area $= \int_{-1}^2 [(x + 2) - x^2] dx = \left[\frac{x^2}{2} + 2x - \frac{x^3}{3} \right]_{-1}^2 = \left(2 + 4 - \frac{8}{3} \right) - \left(\frac{1}{2} - 2 + \frac{1}{3} \right) = \frac{10}{3} - \left(-\frac{7}{6} \right) = \frac{9}{2}$.

- 3 Find the area between $y = \sqrt{x}$ and $y = x^2$ on $[0, 1]$.

On $[0, 1]$, $\sqrt{x} \geq x^2$. Area $= \int_0^1 (\sqrt{x} - x^2) dx = \left[\frac{2}{3}x^{3/2} - \frac{x^3}{3} \right]_0^1 = \frac{2}{3} - \frac{1}{3} = \frac{1}{3}$.

Regions requiring horizontal strips

- 4 Find the area of the region bounded by $x = y^2$ and $x = y + 2$, integrating with respect to y .

Intersect: $y^2 = y + 2 \Rightarrow y^2 - y - 2 = 0 \Rightarrow (y - 2)(y + 1) = 0$; $y = -1, 2$. On this interval $y + 2 \geq y^2$. Area $= \int_{-1}^2 [(y + 2) - y^2] dy = \left[\frac{y^2}{2} + 2y - \frac{y^3}{3} \right]_{-1}^2 = \left(2 + 4 - \frac{8}{3} \right) - \left(\frac{1}{2} - 2 + \frac{1}{3} \right) = \frac{9}{2}$.

Regions with multiple intersection points

- 5 Find the total area between $y = \sin x$ and $y = \cos x$ on $[0, \pi]$, accounting for the curves crossing at $x = \frac{\pi}{4}$.

On $\left[0, \frac{\pi}{4}\right]$, $\cos x > \sin x$; on $\left[\frac{\pi}{4}, \pi\right]$, $\sin x > \cos x$. Total area $= \int_0^{\pi/4} (\cos x - \sin x) dx + \int_{\pi/4}^{\pi} (\sin x - \cos x) dx$. First: $[\sin x + \cos x]_0^{\pi/4} = \sqrt{2} - 1$. Second: $[-\cos x - \sin x]_{\pi/4}^{\pi} = (1 - 0) - (-\sqrt{2}) = 1 + \sqrt{2}$. Total $= (\sqrt{2} - 1) + (1 + \sqrt{2}) = 2\sqrt{2} \approx 2.83$.

Area using the more efficient variable of integration

- 6 Find the area enclosed by $y^2 = 4 - x$ and $x = 0$, choosing the more convenient variable of integration.

Solving for x : $x = 4 - y^2$, bounded by $x = 0$ (i.e. $y = \pm 2$). Integrating with respect to y is easier: Area $= \int_{-2}^2 (4 - y^2) dy = \left[4y - \frac{y^3}{3} \right]_{-2}^2 = \left(8 - \frac{8}{3} \right) - \left(-8 + \frac{8}{3} \right) = \frac{16}{3} + \frac{16}{3} = \frac{32}{3}$.

Setting up but not evaluating (AP-style)

- 7 Set up (but do not evaluate) the integral expression for the area of the region enclosed by $y = e^x$, $y = \ln x$, and the lines $x = 1$ and $x = 2$.

On $[1, 2]$, $e^x \geq \ln x$ (since e^x grows much faster). Area $= \int_1^2 (e^x - \ln x) dx$.

- 8 Set up (but do not evaluate) the integral expression for the area between $y = x^3 - 4x$ and the x -axis on $[-2, 2]$, accounting for the curve crossing the axis at $x = -2, 0, 2$.

On $[-2, 0]$, $x^3 - 4x \geq 0$; on $[0, 2]$, $x^3 - 4x \leq 0$. Total area
 $= \int_{-2}^0 (x^3 - 4x) dx + \int_0^2 -(x^3 - 4x) dx = \int_{-2}^0 (x^3 - 4x) dx - \int_0^2 (x^3 - 4x) dx$.

Area between curves and a table of values

- 9 Functions f and g satisfy $f(x) \geq g(x)$ for all x in $[0, 6]$, with selected values given: $f(0) = 10$, $f(2) = 8$, $f(4) = 9$, $f(6) = 12$ and $g(0) = 3$, $g(2) = 4$, $g(4) = 5$, $g(6) = 6$. Use the Trapezoidal Rule with these 3 equal subintervals to approximate the area between the curves, $\int_0^6 [f(x) - g(x)] dx$.

Let $h(x) = f(x) - g(x)$: $h(0) = 7$, $h(2) = 4$, $h(4) = 4$, $h(6) = 6$. Trapezoidal Rule with width 2:
 $T = 2 \left[\frac{7+4}{2} + \frac{4+4}{2} + \frac{4+6}{2} \right] = 2[5.5 + 4 + 5] = 2(14.5) = 29$.

Visualizing area between two curves

- 10 The graph shows $y = x^2$ and $y = x + 2$ from an earlier problem, together with the region between them.

The line $y = x + 2$ lies above the parabola $y = x^2$ throughout $(-1, 2)$, matching the earlier setup
 $\int_{-1}^2 [(x + 2) - x^2] dx = \frac{9}{2}$.

A capstone area problem

- 11 Find the area of the region bounded by $y = x^3 - x$ and $y = 0$ on $[-1, 1]$, accounting for the sign change at $x = 0$.

On $[-1, 0]$, $x^3 - x \geq 0$; on $[0, 1]$, $x^3 - x \leq 0$. Area $= \int_{-1}^0 (x^3 - x) dx + \left| \int_0^1 (x^3 - x) dx \right|$.
 $\int_{-1}^0 (x^3 - x) dx = \left[\frac{x^4}{4} - \frac{x^2}{2} \right]_{-1}^0 = 0 - \left(\frac{1}{4} - \frac{1}{2} \right) = \frac{1}{4}$. By symmetry the second piece is also $\frac{1}{4}$. Total area $= \frac{1}{2}$.

Area between an exponential curve and a line

- 12 Find the area between $y = e^x$ and $y = x + 1$ on $[0, 2]$ (note $e^x \geq x + 1$ for all x , tangent only at $x = 0$).

Area $= \int_0^2 [e^x - (x + 1)] dx = \left[e^x - \frac{x^2}{2} - x \right]_0^2 = (e^2 - 2 - 2) - (1 - 0 - 0) = e^2 - 5$.

An applied area-between-curves problem

- 13** Two competing companies' profit rates (in thousands of dollars per year) are $P_1(t) = 50 - 2t$ and $P_2(t) = 30 + t$ over the first 10 years. Find the total difference in cumulative profit between the two companies over this period, using the area between the curves.

Intersection: $50 - 2t = 30 + t \Rightarrow 20 = 3t \Rightarrow t = \frac{20}{3} \approx 6.67$. For $t < \frac{20}{3}$, $P_1 > P_2$; for $t > \frac{20}{3}$, $P_2 > P_1$. Total absolute difference = $\left| \int_0^{20/3} (P_1 - P_2) dt \right| + \left| \int_{20/3}^{10} (P_2 - P_1) dt \right|$, which can be computed piecewise using $P_1 - P_2 = 20 - 3t$.

Multiple choice — easy 15

- 14** Find the area between $y = x^2$ and $y = -8x - 12$.

- a. $\frac{35}{3}$
- b. $\frac{32}{3}$ ✓
- c. 8
- d. $\frac{64}{3}$

Intersect: $x^2 = -8x - 12$ gives $x = -6, -2$. Area = $\int_{-6}^{-2} [(-8x - 12) - x^2] dx = \frac{(-2 - (-6))^3}{6} = \frac{4^3}{6} = \frac{32}{3}$.

- 15** Find the area between $y = x^2$ and $y = 7x - 10$.

- a. 9
- b. 5
- c. $\frac{9}{2}$ ✓
- d. $\frac{11}{2}$

Intersect: $x^2 = 7x - 10$ gives $x = 2, 5$. Area = $\int_2^5 [(7x - 10) - x^2] dx = \frac{(5 - (2))^3}{6} = \frac{3^3}{6} = \frac{9}{2}$.

16 Find the area between $y = x^2$ and $y = 10x - 16$.

- a. 37
- b. 72
- c. 18
- d. 36 ✓

Intersect: $x^2 = 10x - 16$ gives $x = 2, 8$. Area $= \int_2^8 [(10x - 16) - x^2] dx = \frac{(8 - (2))^3}{6} = \frac{6^3}{6} = 36$.

17 Find the area between $y = x^2$ and $y = -x + 6$.

- a. $\frac{25}{2}$
- b. $\frac{131}{6}$
- c. $\frac{125}{3}$
- d. $\frac{125}{6}$ ✓

Intersect: $x^2 = -x + 6$ gives $x = -3, 2$. Area $= \int_{-3}^2 [(-x + 6) - x^2] dx = \frac{(2 - (-3))^3}{6} = \frac{5^3}{6} = \frac{125}{6}$.

18 Find the area between $y = x^2$ and $y = -4x - 3$.

- a. $\frac{4}{3}$ ✓
- b. 2
- c. $\frac{8}{3}$
- d. $\frac{7}{3}$

Intersect: $x^2 = -4x - 3$ gives $x = -3, -1$. Area $= \int_{-3}^{-1} [(-4x - 3) - x^2] dx = \frac{(-1 - (-3))^3}{6} = \frac{2^3}{6} = \frac{4}{3}$.

19 Find the area between $y = x^2$ and $y = -2x$.

- a. 2
- b. $\frac{4}{3}$ ✓
- c. $\frac{8}{3}$
- d. $\frac{7}{3}$

Intersect: $x^2 = -2x$ gives $x = -2, 0$. Area $= \int_{-2}^0 [(-2x) - x^2] dx = \frac{(0 - (-2))^3}{6} = \frac{2^3}{6} = \frac{4}{3}$.

20 Find the area between $y = x^2$ and $y = 7x - 6$.

a. $\frac{25}{2}$

b. $\frac{131}{6}$

c. $\frac{125}{6}$ ✓

d. $\frac{125}{3}$

Intersect: $x^2 = 7x - 6$ gives $x = 1, 6$. Area $= \int_1^6 [(7x - 6) - x^2] dx = \frac{(6 - (1))^3}{6} = \frac{5^3}{6} = \frac{125}{6}$.

21 Find the area between $y = x^2$ and $y = 3x$.

a. 9

b. $\frac{11}{2}$

c. $\frac{9}{2}$ ✓

d. 4

Intersect: $x^2 = 3x$ gives $x = 0, 3$. Area $= \int_0^3 [(3x) - x^2] dx = \frac{(3 - (0))^3}{6} = \frac{3^3}{6} = \frac{9}{2}$.

22 Find the area between $y = x^2$ and $y = 4x + 5$.

a. 72

b. 36 ✓

c. 37

d. 18

Intersect: $x^2 = 4x + 5$ gives $x = -1, 5$. Area $= \int_{-1}^5 [(4x + 5) - x^2] dx = \frac{(5 - (-1))^3}{6} = \frac{6^3}{6} = 36$.

23 Find the area between $y = x^2$ and $y = 5x - 6$.

- a. $\frac{1}{2}$
- b. $\frac{1}{6}$ ✓
- c. $\frac{1}{3}$
- d. $\frac{7}{6}$

Intersect: $x^2 = 5x - 6$ gives $x = 2, 3$. Area $= \int_2^3 [(5x - 6) - x^2] dx = \frac{(3 - (2))^3}{6} = \frac{1^3}{6} = \frac{1}{6}$.

24 Find the area between $y = x^2$ and $y = -4x + 5$.

- a. 18
- b. 37
- c. 36 ✓
- d. 72

Intersect: $x^2 = -4x + 5$ gives $x = -5, 1$. Area $= \int_{-5}^1 [(-4x + 5) - x^2] dx = \frac{(1 - (-5))^3}{6} = \frac{6^3}{6} = 36$.

25 Find the area between $y = x^2$ and $y = x + 2$.

- a. $\frac{9}{2}$ ✓
- b. 9
- c. $\frac{11}{2}$
- d. 5

Intersect: $x^2 = x + 2$ gives $x = -1, 2$. Area $= \int_{-1}^2 [(x + 2) - x^2] dx = \frac{(2 - (-1))^3}{6} = \frac{3^3}{6} = \frac{9}{2}$.

26 Find the area between $y = x^2$ and $y = 6x$.

- a. 72
- b. 18
- c. 37
- d. 36 ✓

Intersect: $x^2 = 6x$ gives $x = 0, 6$. Area $= \int_0^6 [(6x) - x^2] dx = \frac{(6 - (0))^3}{6} = \frac{6^3}{6} = 36$.

27 Find the area between $y = x^2$ and $y = -10x - 24$.

- a. $\frac{8}{3}$
- b. $\frac{7}{3}$
- c. 2
- d. $\frac{4}{3}$ ✓

Intersect: $x^2 = -10x - 24$ gives $x = -6, -4$. Area $= \int_{-6}^{-4} [(-10x - 24) - x^2] dx = \frac{(-4 - (-6))^3}{6} = \frac{2^3}{6} = \frac{4}{3}$.

28 Find the area between $y = x^2$ and $y = 9x - 14$.

- a. $\frac{25}{2}$
- b. $\frac{125}{6}$ ✓
- c. $\frac{131}{6}$
- d. $\frac{125}{3}$

Intersect: $x^2 = 9x - 14$ gives $x = 2, 7$. Area $= \int_2^7 [(9x - 14) - x^2] dx = \frac{(7 - (2))^3}{6} = \frac{5^3}{6} = \frac{125}{6}$.

Multiple choice — medium 15

29 Find the area between $y = x^2$ and $y = x^7$ on $[0, 1]$.

- a. $\frac{5}{24}$ ✓
- b. $\frac{11}{24}$
- c. $\frac{5}{14}$
- d. $-\frac{5}{24}$

On $[0, 1]$, $x^2 \geq x^7$. Area $= \int_0^1 (x^2 - x^7) dx = \left[\frac{x^3}{3} - \frac{x^8}{8} \right]_0^1 = \frac{1}{3} - \frac{1}{8} = \frac{5}{24}$.

30 Find the area between $y = x^4$ and $y = x^9$ on $[0, 1]$.

a. $-\frac{1}{10}$

b. $\frac{3}{10}$

c. $\frac{1}{10}$ ✓

d. $\frac{5}{36}$

On $[0, 1]$, $x^4 \geq x^9$. Area $= \int_0^1 (x^4 - x^9) dx = \left[\frac{x^5}{5} - \frac{x^{10}}{10} \right]_0^1 = \frac{1}{5} - \frac{1}{10} = \frac{1}{10}$.

31 Find the area between $y = x$ and $y = x^6$ on $[0, 1]$.

a. $\frac{5}{6}$

b. $\frac{9}{14}$

c. $\frac{5}{14}$ ✓

d. $-\frac{5}{14}$

On $[0, 1]$, $x \geq x^6$. Area $= \int_0^1 (x - x^6) dx = \left[\frac{x^2}{2} - \frac{x^7}{7} \right]_0^1 = \frac{1}{2} - \frac{1}{7} = \frac{5}{14}$.

32 Find the area between $y = x^2$ and $y = x^3$ on $[0, 1]$.

a. $\frac{1}{12}$ ✓

b. $\frac{7}{12}$

c. $\frac{1}{6}$

d. $-\frac{1}{12}$

On $[0, 1]$, $x^2 \geq x^3$. Area $= \int_0^1 (x^2 - x^3) dx = \left[\frac{x^3}{3} - \frac{x^4}{4} \right]_0^1 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12}$.

33 Find the area between $y = x$ and $y = x^5$ on $[0, 1]$.

a. $\frac{1}{3}$ ✓

b. $-\frac{1}{3}$

c. $\frac{2}{3}$

d. $\frac{4}{5}$

On $[0, 1]$, $x \geq x^5$. Area $= \int_0^1 (x - x^5) dx = \left[\frac{x^2}{2} - \frac{x^6}{6} \right]_0^1 = \frac{1}{2} - \frac{1}{6} = \frac{1}{3}$.

34 Find the area between $y = x^3$ and $y = x^4$ on $[0, 1]$.

a. $\frac{9}{20}$

b. $-\frac{1}{20}$

c. $\frac{1}{12}$

d. $\frac{1}{20}$ ✓

On $[0, 1]$, $x^3 \geq x^4$. Area $= \int_0^1 (x^3 - x^4) dx = \left[\frac{x^4}{4} - \frac{x^5}{5} \right]_0^1 = \frac{1}{4} - \frac{1}{5} = \frac{1}{20}$.

35 Find the area between $y = x^4$ and $y = x^6$ on $[0, 1]$.

a. $\frac{12}{35}$

b. $-\frac{2}{35}$

c. $\frac{1}{12}$

d. $\frac{2}{35}$ ✓

On $[0, 1]$, $x^4 \geq x^6$. Area $= \int_0^1 (x^4 - x^6) dx = \left[\frac{x^5}{5} - \frac{x^7}{7} \right]_0^1 = \frac{1}{5} - \frac{1}{7} = \frac{2}{35}$.

36 Find the area between $y = x^4$ and $y = x^5$ on $[0, 1]$.

a. $\frac{1}{30}$ ✓

b. $-\frac{1}{30}$

c. $\frac{1}{20}$

d. $\frac{11}{30}$

On $[0, 1]$, $x^4 \geq x^5$. Area $= \int_0^1 (x^4 - x^5) dx = \left[\frac{x^5}{5} - \frac{x^6}{6} \right]_0^1 = \frac{1}{5} - \frac{1}{6} = \frac{1}{30}$.

37 Find the area between $y = x^2$ and $y = x^4$ on $[0, 1]$.

- a. $\frac{8}{15}$
- b. $\frac{2}{15}$ ✓
- c. $\frac{1}{4}$
- d. $-\frac{2}{15}$

On $[0, 1]$, $x^2 \geq x^4$. Area $= \int_0^1 (x^2 - x^4) dx = \left[\frac{x^3}{3} - \frac{x^5}{5} \right]_0^1 = \frac{1}{3} - \frac{1}{5} = \frac{2}{15}$.

38 Find the area between $y = x^3$ and $y = x^5$ on $[0, 1]$.

- a. $\frac{1}{12}$ ✓
- b. $\frac{5}{12}$
- c. $\frac{2}{15}$
- d. $-\frac{1}{12}$

On $[0, 1]$, $x^3 \geq x^5$. Area $= \int_0^1 (x^3 - x^5) dx = \left[\frac{x^4}{4} - \frac{x^6}{6} \right]_0^1 = \frac{1}{4} - \frac{1}{6} = \frac{1}{12}$.

39 Find the area between $y = x^3$ and $y = x^6$ on $[0, 1]$.

- a. $\frac{11}{28}$
- b. $\frac{1}{6}$
- c. $\frac{3}{28}$ ✓
- d. $-\frac{3}{28}$

On $[0, 1]$, $x^3 \geq x^6$. Area $= \int_0^1 (x^3 - x^6) dx = \left[\frac{x^4}{4} - \frac{x^7}{7} \right]_0^1 = \frac{1}{4} - \frac{1}{7} = \frac{3}{28}$.

40 Find the area between $y = x$ and $y = x^3$ on $[0, 1]$.

- a. $-\frac{1}{4}$
- b. $\frac{3}{4}$
- c. $\frac{2}{3}$
- d. $\frac{1}{4}$ ✓

On $[0, 1]$, $x \geq x^3$. Area $= \int_0^1 (x - x^3) dx = \left[\frac{x^2}{2} - \frac{x^4}{4} \right]_0^1 = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$.

41 Find the area between $y = x$ and $y = x^4$ on $[0, 1]$.

a. $\frac{3}{10}$ ✓

b. $\frac{3}{4}$

c. $\frac{7}{10}$

d. $-\frac{3}{10}$

On $[0, 1]$, $x \geq x^4$. Area $= \int_0^1 (x - x^4) dx = \left[\frac{x^2}{2} - \frac{x^5}{5} \right]_0^1 = \frac{1}{2} - \frac{1}{5} = \frac{3}{10}$.

42 Find the area between $y = x^4$ and $y = x^7$ on $[0, 1]$.

a. $\frac{13}{40}$

b. $-\frac{3}{40}$

c. $\frac{3}{28}$

d. $\frac{3}{40}$ ✓

On $[0, 1]$, $x^4 \geq x^7$. Area $= \int_0^1 (x^4 - x^7) dx = \left[\frac{x^5}{5} - \frac{x^8}{8} \right]_0^1 = \frac{1}{5} - \frac{1}{8} = \frac{3}{40}$.

43 Find the area between $y = x^2$ and $y = x^5$ on $[0, 1]$.

a. $\frac{1}{2}$

b. $-\frac{1}{6}$

c. $\frac{1}{6}$ ✓

d. $\frac{3}{10}$

On $[0, 1]$, $x^2 \geq x^5$. Area $= \int_0^1 (x^2 - x^5) dx = \left[\frac{x^3}{3} - \frac{x^6}{6} \right]_0^1 = \frac{1}{3} - \frac{1}{6} = \frac{1}{6}$.

Multiple choice — hard 15

- 44 Find the total area between $y = x^3 - 36x$ and the x -axis on $[-6, 6]$, accounting for the sign change at $x = -6, 0, 6$.
- a. 324
- b. 654
- c. 108
- d. 648 ✓

By symmetry, total area $= 2 \int_0^6 (6^2x - x^3) dx = 2 \left[\frac{6^2x^2}{2} - \frac{x^4}{4} \right]_0^6 = 2 \left(\frac{1296}{2} - \frac{1296}{4} \right) = 2 \left(\frac{1296}{4} \right) = \frac{1296}{2} = 648$.

- 45 Find the total area between $y = x^3 - 9x$ and the x -axis on $[-3, 3]$, accounting for the sign change at $x = -3, 0, 3$.
- a. $\frac{87}{2}$
- b. $\frac{81}{2}$ ✓
- c. $\frac{27}{2}$
- d. $\frac{81}{4}$

By symmetry, total area $= 2 \int_0^3 (3^2x - x^3) dx = 2 \left[\frac{3^2x^2}{2} - \frac{x^4}{4} \right]_0^3 = 2 \left(\frac{81}{2} - \frac{81}{4} \right) = 2 \left(\frac{81}{4} \right) = \frac{81}{2} = \frac{81}{2}$.

- 46 Find the total area between $y = x^3 - 16x$ and the x -axis on $[-4, 4]$, accounting for the sign change at $x = -4, 0, 4$.
- a. 132
- b. 64
- c. 128 ✓
- d. 32

By symmetry, total area $= 2 \int_0^4 (4^2x - x^3) dx = 2 \left[\frac{4^2x^2}{2} - \frac{x^4}{4} \right]_0^4 = 2 \left(\frac{256}{2} - \frac{256}{4} \right) = 2 \left(\frac{256}{4} \right) = \frac{256}{2} = 128$.

- 47 Find the total area between $y = x^3 - 4x$ and the x -axis on $[-2, 2]$, accounting for the sign change at $x = -2, 0, 2$.

- a. 4
- b. $\frac{17}{2}$
- c. 10
- d. 8 ✓

By symmetry, total area $= 2 \int_0^2 (2^2x - x^3) dx = 2 \left[\frac{2^2x^2}{2} - \frac{x^4}{4} \right]_0^2 = 2 \left(\frac{16}{2} - \frac{16}{4} \right) = 2 \left(\frac{16}{4} \right) = \frac{16}{2} = 8$.

- 48 Find the total area between $y = x^3 - 25x$ and the x -axis on $[-5, 5]$, accounting for the sign change at $x = -5, 0, 5$.

- a. $\frac{635}{2}$
- b. $\frac{125}{2}$
- c. $\frac{625}{2}$ ✓
- d. $\frac{625}{4}$

By symmetry, total area $= 2 \int_0^5 (5^2x - x^3) dx = 2 \left[\frac{5^2x^2}{2} - \frac{x^4}{4} \right]_0^5 = 2 \left(\frac{625}{2} - \frac{625}{4} \right) = 2 \left(\frac{625}{4} \right) = \frac{625}{2} = \frac{625}{2}$.

- 49 Functions f and g satisfy $f(x) \geq g(x)$ on $[0, 3]$, with $h(x) = f(x) - g(x)$ given by $h(0) = 8$, $h(1) = 15$, $h(2) = 6$, $h(3) = 10$. Use the Trapezoidal Rule with these 3 equal subintervals (width 1) to approximate the area between the curves.

- a. 39
- b. 29
- c. 31
- d. 30 ✓

$T = 1 \left[\frac{8+15}{2} + \frac{15+6}{2} + \frac{6+10}{2} \right] = 1 \left[\frac{8+10}{2} + 15 + 6 \right] = 1(9 + 15 + 6) = 30$.

- 50 Functions f and g satisfy $f(x) \geq g(x)$ on $[0, 12]$, with $h(x) = f(x) - g(x)$ given by $h(0) = 8$, $h(4) = 9$, $h(8) = 14$, $h(12) = 2$. Use the Trapezoidal Rule with these 3 equal subintervals (width 4) to approximate the area between the curves.

a. 112 ✓

b. 132

c. 116

d. 108

$$T = 4 \left[\frac{8+9}{2} + \frac{9+14}{2} + \frac{14+2}{2} \right] = 4 \left[\frac{8+2}{2} + 9 + 14 \right] = 4(5 + 9 + 14) = 112.$$

- 51 Functions f and g satisfy $f(x) \geq g(x)$ on $[0, 6]$, with $h(x) = f(x) - g(x)$ given by $h(0) = 2$, $h(2) = 12$, $h(4) = 13$, $h(6) = 2$. Use the Trapezoidal Rule with these 3 equal subintervals (width 2) to approximate the area between the curves.

a. 58

b. 56

c. 52

d. 54 ✓

$$T = 2 \left[\frac{2+12}{2} + \frac{12+13}{2} + \frac{13+2}{2} \right] = 2 \left[\frac{2+2}{2} + 12 + 13 \right] = 2(2 + 12 + 13) = 54.$$

- 52 Functions f and g satisfy $f(x) \geq g(x)$ on $[0, 3]$, with $h(x) = f(x) - g(x)$ given by $h(0) = 8$, $h(1) = 14$, $h(2) = 6$, $h(3) = 6$. Use the Trapezoidal Rule with these 3 equal subintervals (width 1) to approximate the area between the curves.

a. 26

b. 34

c. 28

d. 27 ✓

$$T = 1 \left[\frac{8+14}{2} + \frac{14+6}{2} + \frac{6+6}{2} \right] = 1 \left[\frac{8+6}{2} + 14 + 6 \right] = 1(7 + 14 + 6) = 27.$$

- 53 Functions f and g satisfy $f(x) \geq g(x)$ on $[0, 12]$, with $h(x) = f(x) - g(x)$ given by $h(0) = 10$, $h(4) = 15$, $h(8) = 8$, $h(12) = 8$. Use the Trapezoidal Rule with these 3 equal subintervals (width 4) to approximate the area between the curves.

a. 128 ✓

b. 132

c. 124

d. 164

$$T = 4 \left[\frac{10+15}{2} + \frac{15+8}{2} + \frac{8+8}{2} \right] = 4 \left[\frac{10+8}{2} + 15 + 8 \right] = 4(9 + 15 + 8) = 128.$$

- 54 Find the area between $y = e^{-2x}$ and $y = -2x + 1$ on $[0, 1]$ (note $e^{-2x} \geq -2x + 1$ for all x , tangent only at $x = 0$), rounded to two decimal places.

a. 0.43 ✓

b. 0.52

c. 0.35

d. 1.43

$$\text{Area} = \int_0^1 [e^{-2x} - (-2x + 1)] dx = \left[\frac{e^{-2x}}{-2} - \frac{-2x^2}{2} - x \right]_0^1 \approx 0.43.$$

- 55 Find the area between $y = e^{2x}$ and $y = 2x + 1$ on $[0, 2]$ (note $e^{2x} \geq 2x + 1$ for all x , tangent only at $x = 0$), rounded to two decimal places.

a. 21.8

b. 24.96

c. 16.64

d. 20.8 ✓

$$\text{Area} = \int_0^2 [e^{2x} - (2x + 1)] dx = \left[\frac{e^{2x}}{2} - \frac{2x^2}{2} - x \right]_0^2 \approx 20.8.$$

- 56 Find the area between $y = e^{-x}$ and $y = -x + 1$ on $[0, 3]$ (note $e^{-x} \geq -x + 1$ for all x , tangent only at $x = 0$), rounded to two decimal places.

a. 1.96

b. 2.45 ✓

c. 2.94

d. 3.45

$$\text{Area} = \int_0^3 [e^{-x} - (-x + 1)] dx = \left[\frac{e^{-x}}{-1} - \frac{-x^2}{2} - x \right]_0^3 \approx 2.45.$$

- 57 Find the area between $y = e^{2x}$ and $y = 2x + 1$ on $[0, 1]$ (note $e^{2x} \geq 2x + 1$ for all x , tangent only at $x = 0$), rounded to two decimal places.

a. 2.19

b. 1.43

c. 0.96

d. 1.19 ✓

$$\text{Area} = \int_0^1 [e^{2x} - (2x + 1)] dx = \left[\frac{e^{2x}}{2} - \frac{2x^2}{2} - x \right]_0^1 \approx 1.19.$$

- 58 Find the area between $y = e^x$ and $y = x + 1$ on $[0, 3]$ (note $e^x \geq x + 1$ for all x , tangent only at $x = 0$), rounded to two decimal places.

a. 12.59

b. 13.9

c. 9.27

d. 11.59 ✓

$$\text{Area} = \int_0^3 [e^x - (x + 1)] dx = \left[\frac{e^x}{1} - \frac{x^2}{2} - x \right]_0^3 \approx 11.59.$$